CS & IT ENGINEERING

Operating System

Memory Management

Lecture -1

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Recap of Previous Lecture

doubt ?

Topic Banker's Algorithm

Topic Deadlock Detection

Topics to be Covered

Topic Memory Management

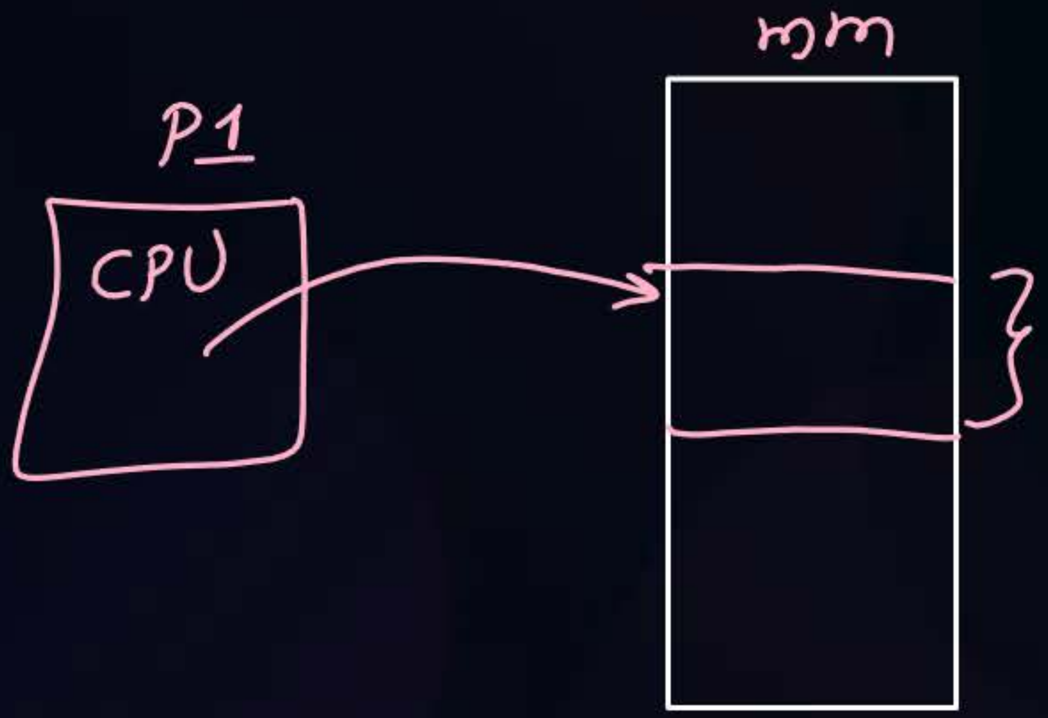
Topic Contiguous Memory Management Technique

Topic : Memory Management

module of OS which manages mm.

Topic : Functions of Memory Management

1. Memory allocation To new process
2. Memory deallocation for terminated process
3. Memory protection ⇒ a process while executing in CPU can access allocated memory of it's own.



Topic : Goals of Memory Management

1. Maximum Utilization of space
2. Ability to run larger programs with limited space

Topic : Memory Management Techniques

Contiguous

fixed partition
(Internal fragmentation)

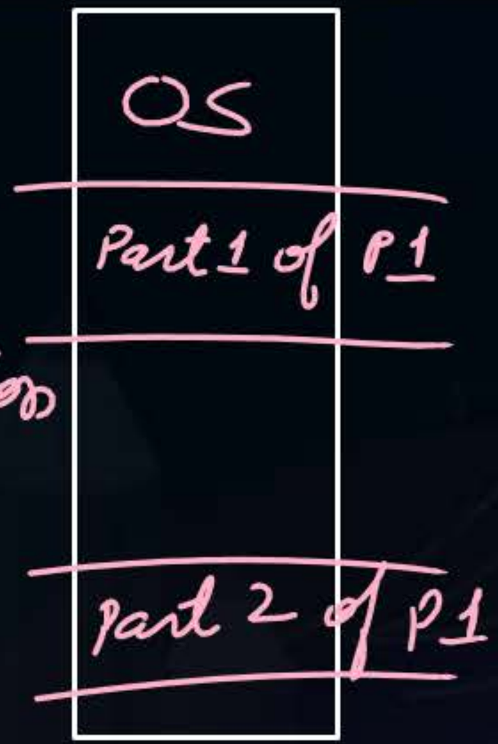
variable partition
(external fragmentation)

Non-contiguous

Process is divided into partitions and the partitions can be stored anywhere in memory.

Paging
⇓
equal size partitions

Segmentation
⇓
variable size partitions



Topic : Contiguous Memory Management

- Entire process should be stored on consecutive memory locations

Topic : Fixed Partition Contiguous MMT

In MMT the mm is divided into fixed no. of partitions (of variable sizes) and each partition is used to store one process.

ex:-

Degree of multiprogramming is limited by number of partitions

It suffers from Internal fragmentation.

↓

the extra space allocated to a process which is more than the required space of process.

Topic : Partition Allocation Policy

which partition should be allocated to a new process.

1. First Fit ⇒ allocate first empty partition which can store process.
2. Best fit ⇒ allocate smallest possible partition which can store process
3. worst fit ⇒ — " — largest — " ——— " ———
4. Next fit ⇒ allocate next empty partition from current allocated partition, which can store process.

Topic : Partition Allocation Policy

4 partitions of size: 100KB, 120KB, 150KB and 80KB

	P1 (Size = 110KB)	P2 (Size = 70KB)	Amount of internal fragmentation
First Fit	120KB	100KB	10 + 30 = 40 KB
Best Fit	120 KB	80 KB	10 + 10 = 20 KB
Worst Fit	150 KB	120 KB	40 + 50 = 90 KB
Next Fit	120 KB	150 KB	10 + 80 = 90 KB

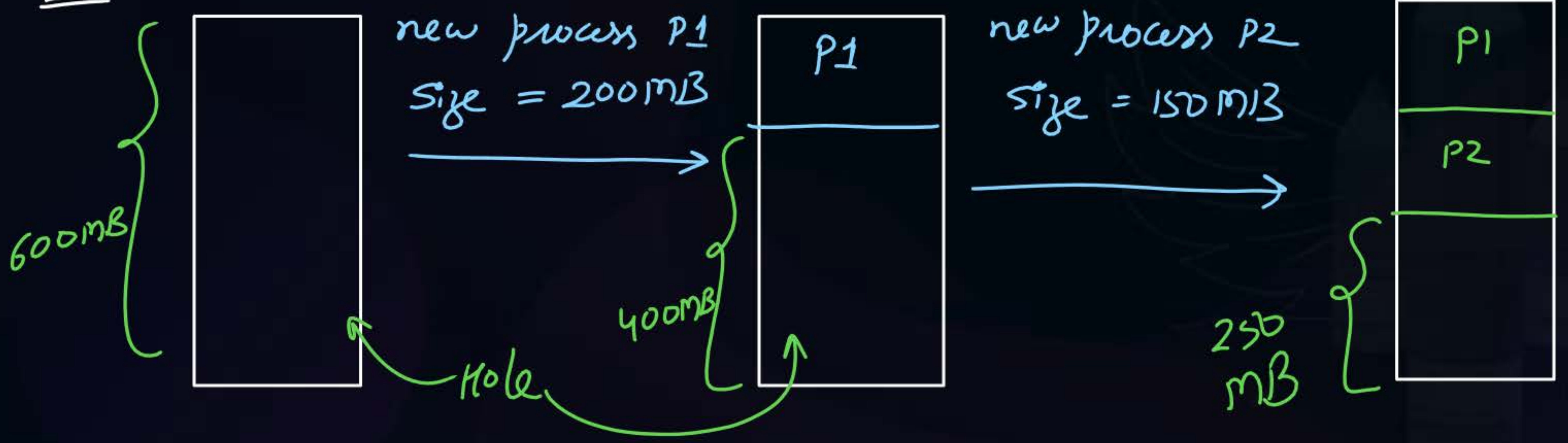
Topic : Variable Partition Contiguous MMT

The mm is not divided into partition priorly.

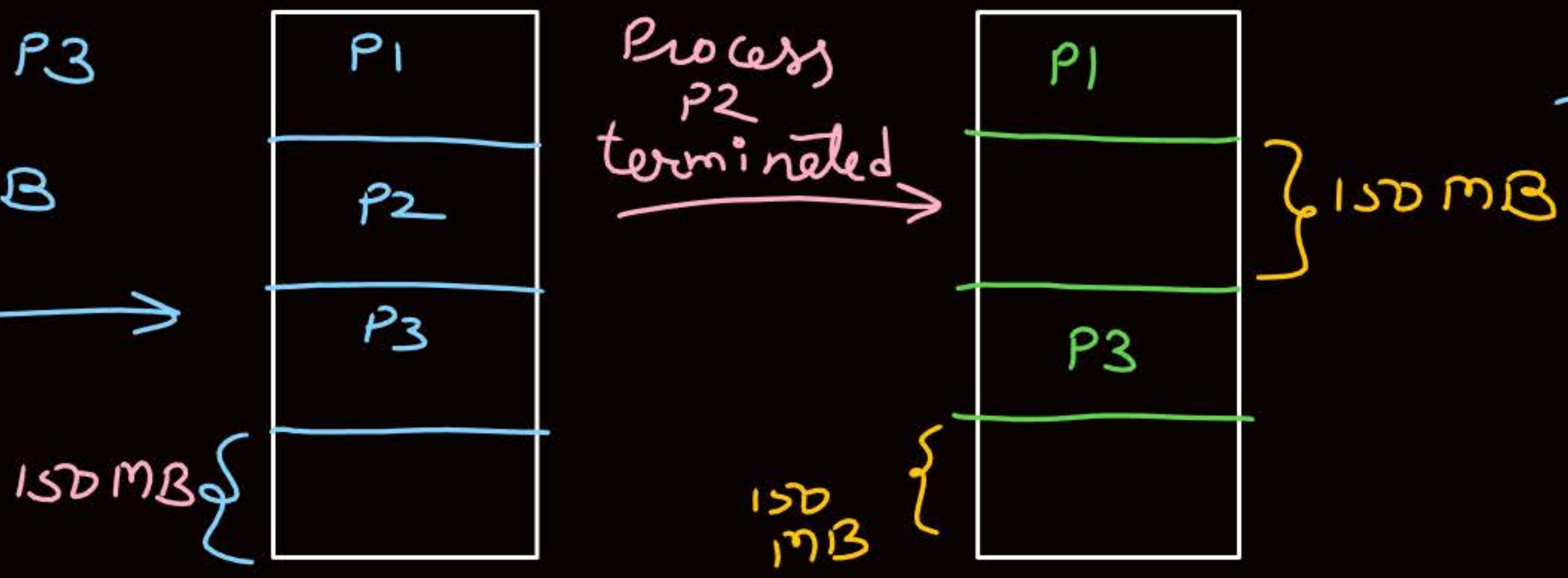
when a new process arrives, then a new partition is created of size of process. And the process is stored in this partition.

⇒ There is no any internal fragmentation.

ex:-



new process P3
size = 100MB

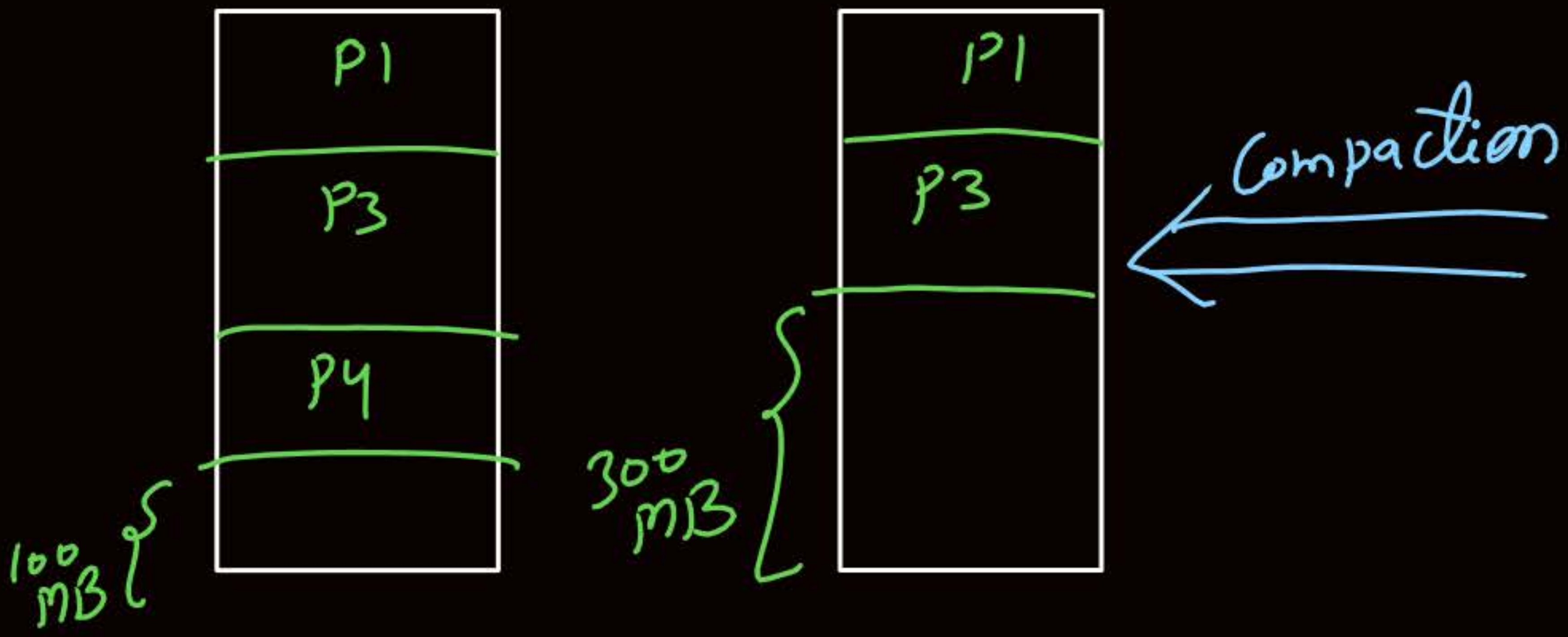


new process P4
size = 200MB

P4 can not be stored.
Because 200MB space is not available consecutively.

⇓

it is called as external fragmentation.



External fragmentation :-

Enough space is available to store a process but not consecutively. Hence process cannot be stored in memory.

Compaction :-

Gather all allocated processes in one side of memory to have all holes (empty spaces) in other side of memory all together to store a new process.

[MCQ]

#Q. Consider the requests from processes in given order 300K, 25K, 125K, and 50K. Let there be two blocks of memory available of size 150K followed by a block size 350K. Which of the following partition allocation schemes can satisfy the above requests? (variable partition)

Best fit

A Best fit but not first fit

B First fit but not best fit

First Fit:-

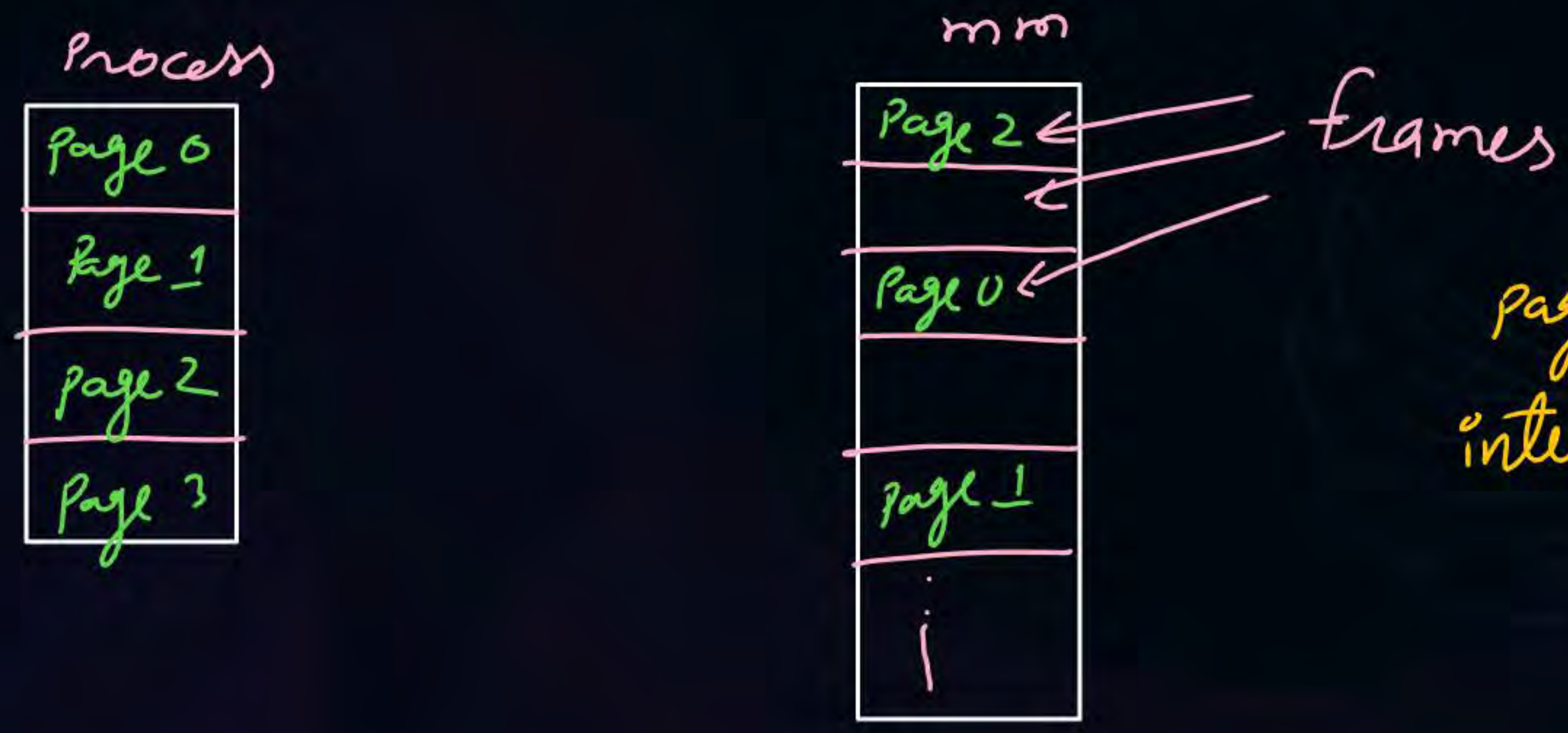
C Both First fit & Best fit

D neither first fit nor best fit

Topic : Paging

The process is divided into equal size partitions called as pages.

The mm is also divided into same size partitions called as frames



paging can suffer from internal fragmentation

2 mins Summary

Topic Memory Management

Topic Contiguous Memory Management Technique

Happy Learning

THANK - YOU